

Q1. Why are smart tweezers used in the industry? Is it worth buying them for a small lab?

Pooja
Through e-mail

A1. A smart tweezer is a highly useful and comfortable handheld LCR meter that is specifically designed to identify and test basic SMD components. SMD components are getting progressively smaller, with some too small to even be labelled. In which case, if they are



Fig. 1: Smart tweezer in use

out from the strip, you never know which component is it. Smart tweezers are highly effective in such cases since they automatically detect the type of component—resistor, capacitor or inductor—and indicate its value with very good accuracy.

The standard LCR meters come in shapes similar to that of a multimeter with cable probes connected to them. These meters are not very comfortable to work with SMD components. You cannot even touch and hold a 0402 package with these probes. But precise gold-plated tweezers or smart tweezers are able to pick and reliably check even the smallest SMD components and take measurements from already mounted devices. (Refer the table for specifications of smart tweezers.)

Q2. I have just started exploring PCB design. I cannot understand what thermal reliefs are used for. Please help me.

Gaurang Kaparwan
Through e-mail

A2. While soldering, when the solder tip touches the joint, its tempera-

ture drops as the heat is dissipated at the joint. If the part you are soldering can dissipate heat at a rate that is higher than what your soldering iron can deliver, the temperature will keep dropping until it won't melt the solder any more. This is especially the case when you have a circuit board with internal planes, large poured planes, or wide conducting traces. Without the relief patterns on copper planes, the planes act as a heat sink drawing much

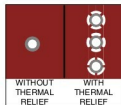


Fig. 2: Thermal relief

of the heat away from the lead you are trying to solder or desolder. The result could be a dry solder joint, or the need to apply excess

heat while soldering. Fig. 2 shows a comparison between a via with and without thermal relief.

Q3. What does record length specification mean in oscilloscopes?

Kaartik Sahaney
Through e-mail

A3. The oscilloscope cannot store an infinite number of samples as assumed by the Nyquist equation. The record length determines the time that can be captured by each channel of the oscilloscope.

Time captured = Record length / Sample rate

Normally, entry-level oscilloscopes come with 2 to 2.5k points, which is more than enough. In general, the greater the record length the better. To capture and debug a serial bus you will need a record length as high as 1M points.

Answers compiled by EFY technical editor, Ankit Gupta. Letters and questions for publication may be addressed to Editor, Electronics For You, D-87/A, Okhla Industrial Area, Phase 1, New Delhi 110020 (E-mail: editsec@efyindia.com) and should include name and address of the sender

Specifications of Smart Tweezers

Smart Tweezers	INDE Enterprises ST-5R
Basic specifications	
Measured parameter	C, L, R, ESR, Rs, Rp
Measuring frequency	100 Hz, 120 Hz, 1 kHz, 10 kHz
Measurement rate	1 time per second, default
DC voltage	No
Resistance range	0.05 ohms to 9.99 mega-ohms
Capacitance range	0.5 pF to 4999 μ F
Inductance range	1 μ H to 999 mH
Accuracy specifications	
Resistance Range	0.1 ohm to 5 mega-ohms
Accuracy	0.2% in range 0.1R - 5M
Test frequency	1 kHz
Capacitance Range	0.5 pF to 4999 μ F
Accuracy	0.5% in range 10 pF - 499 μ F
Test frequency	1 kHz C > 1000 pF; 10 kHz C < 1000 pF; 100 Hz C > 1 μ F
Inductance Range	1 μ H to 1H
Accuracy	0.5% in range 1 μ H - 1H
Test frequency	10 kHz L < 1 μ H; 1 kHz L > 1 μ H; 100 Hz L > 1 mH
Physical specifications	
Operating temperature	0°C to 50°C
Battery type	3.7V LiPo rechargeable 150mAh
Battery life	80 hours for fully-charged battery
Size	14.0x2.5x3.0 cm ³
Weight	53 grams